

Customer Segmentation Analysis

Olist E-Commerce Dataset — RFM-Based Clustering Report

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Executive Summary

This report presents a customer segmentation analysis of 93,357 delivered-order customers from the Olist e-commerce platform, based on RFM (Recency, Frequency, Monetary) features plus review score and delivery time.

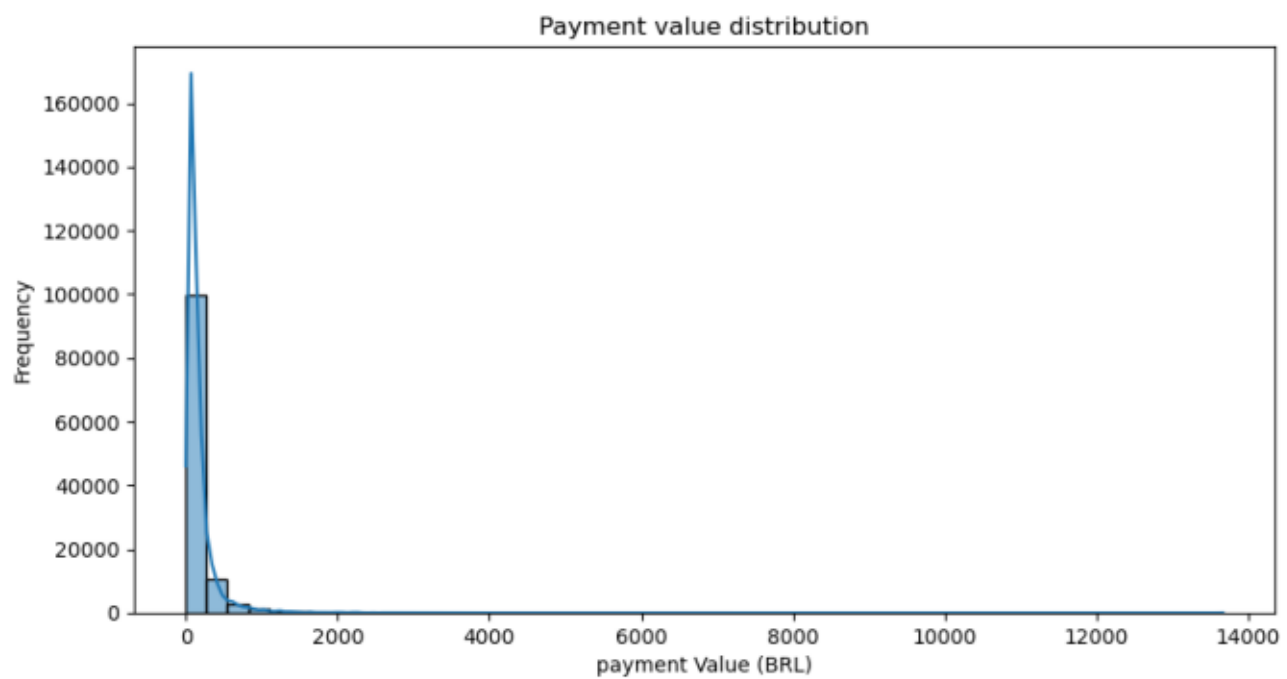
Four segments were identified using KMeans clustering ($k=4$), validated with the Elbow Method and Silhouette Score: High Value, Engaged/Recent, At Risk/Dissatisfied, and Lost/Inactive.

Key Finding: Delivery time is the strongest driver of customer satisfaction, showing a -0.30 correlation with review score. The At Risk/Dissatisfied segment has the slowest average delivery time (17 days) and the lowest review score (1.64), confirming this relationship.

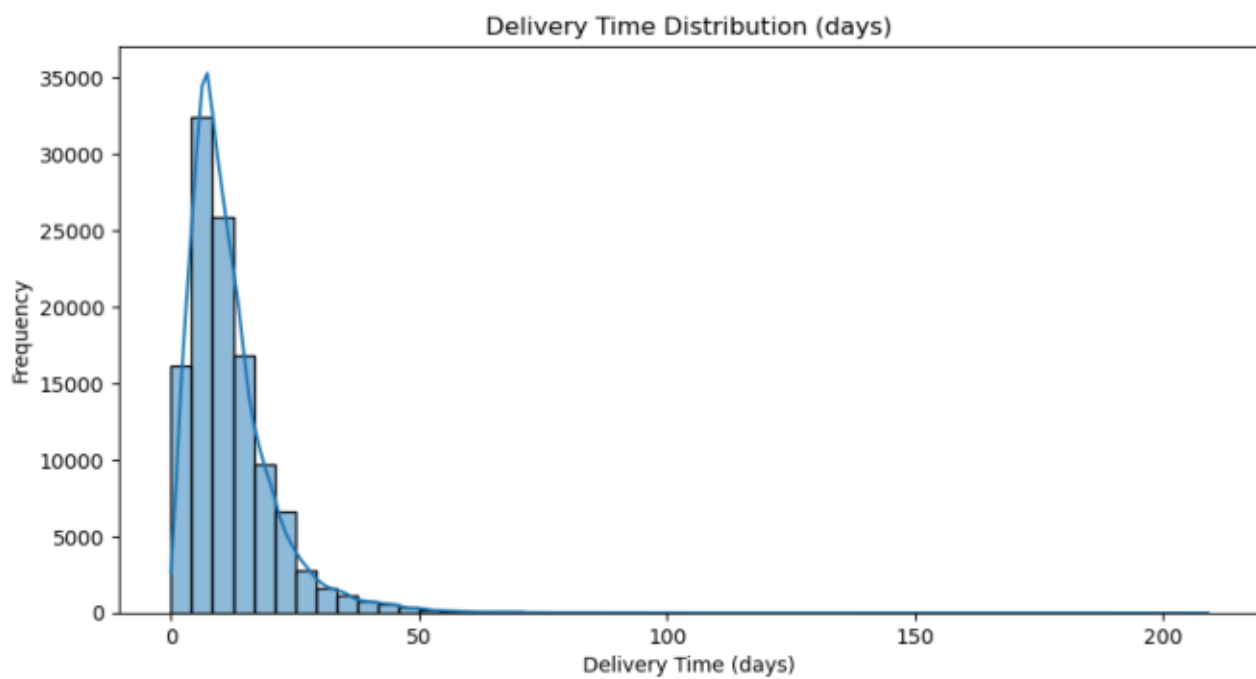
Only 3% of customers make a repeat purchase, highlighting retention as a major growth opportunity.

Recommendations:

1. Improve delivery logistics, prioritizing the At Risk segment.
2. Launch win-back campaigns for the Lost/Inactive segment.
3. Build a loyalty program for the High Value segment.



Payment Value Distribution — right-skewed, most transactions are low value



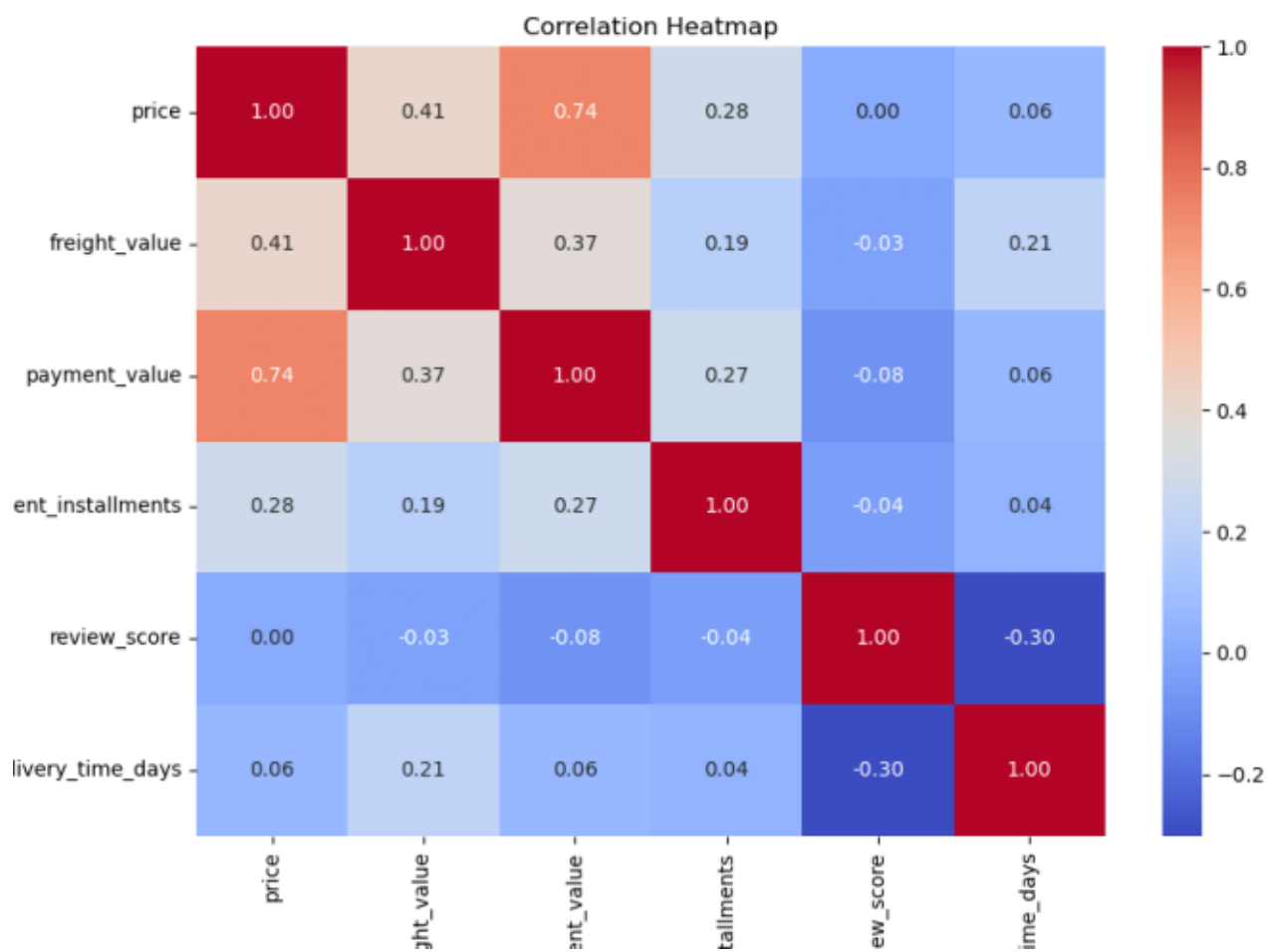
Delivery Time Distribution — most orders deliver within 5-15 days



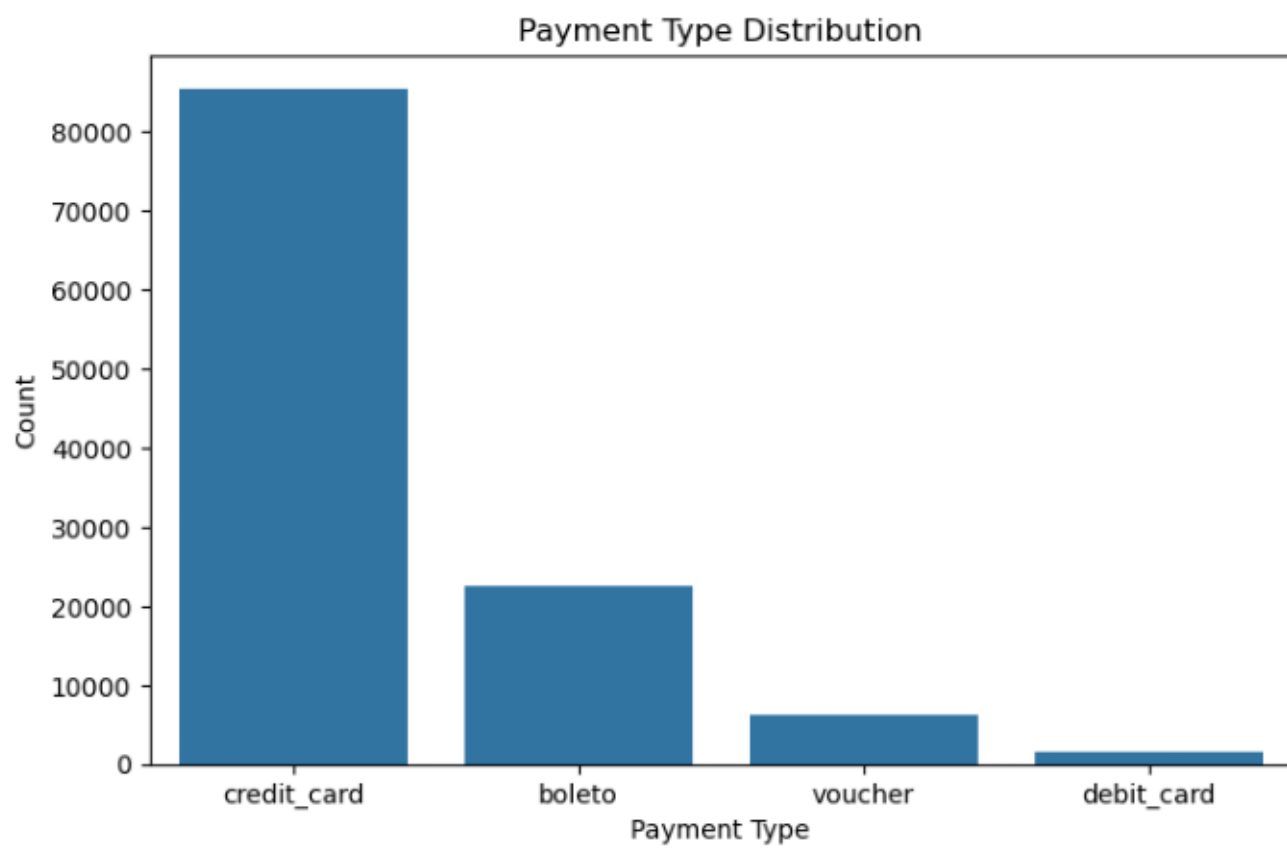
Delivery Time vs Review Score — clear negative relationship



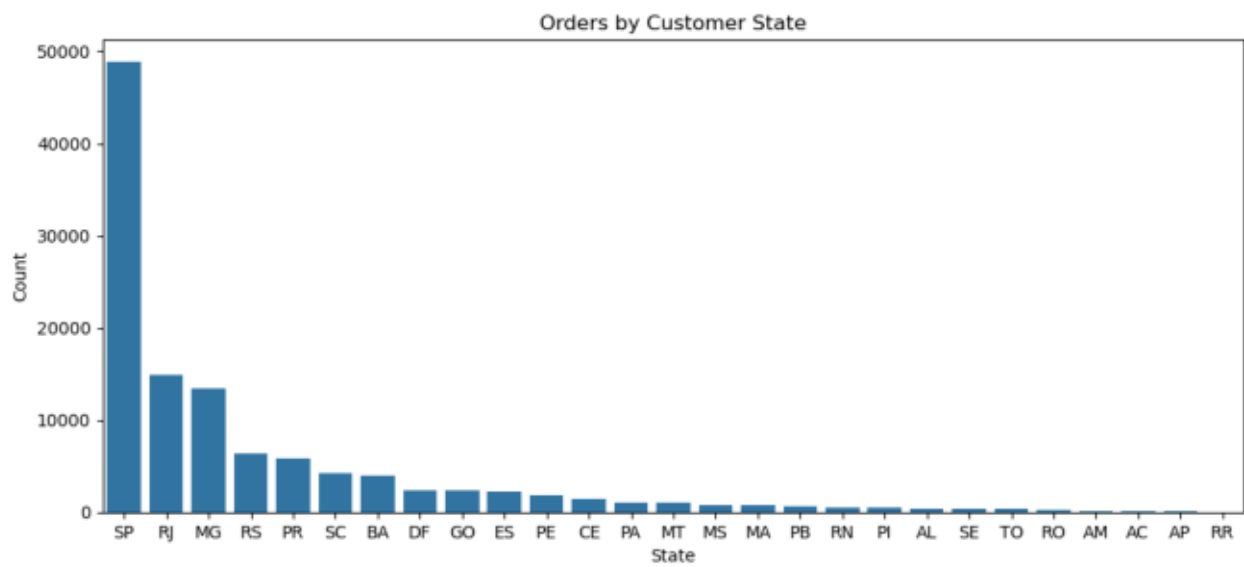
Monthly Orders Trend — steady growth with a seasonal spike



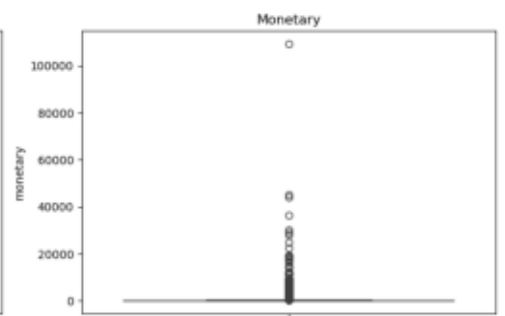
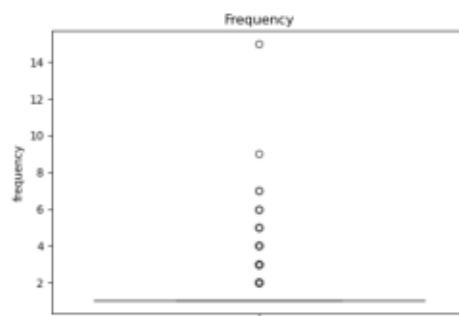
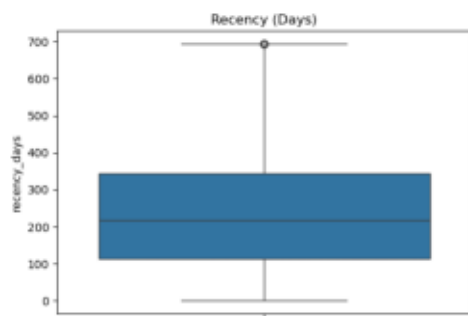
Correlation Heatmap — delivery time most correlated with review score



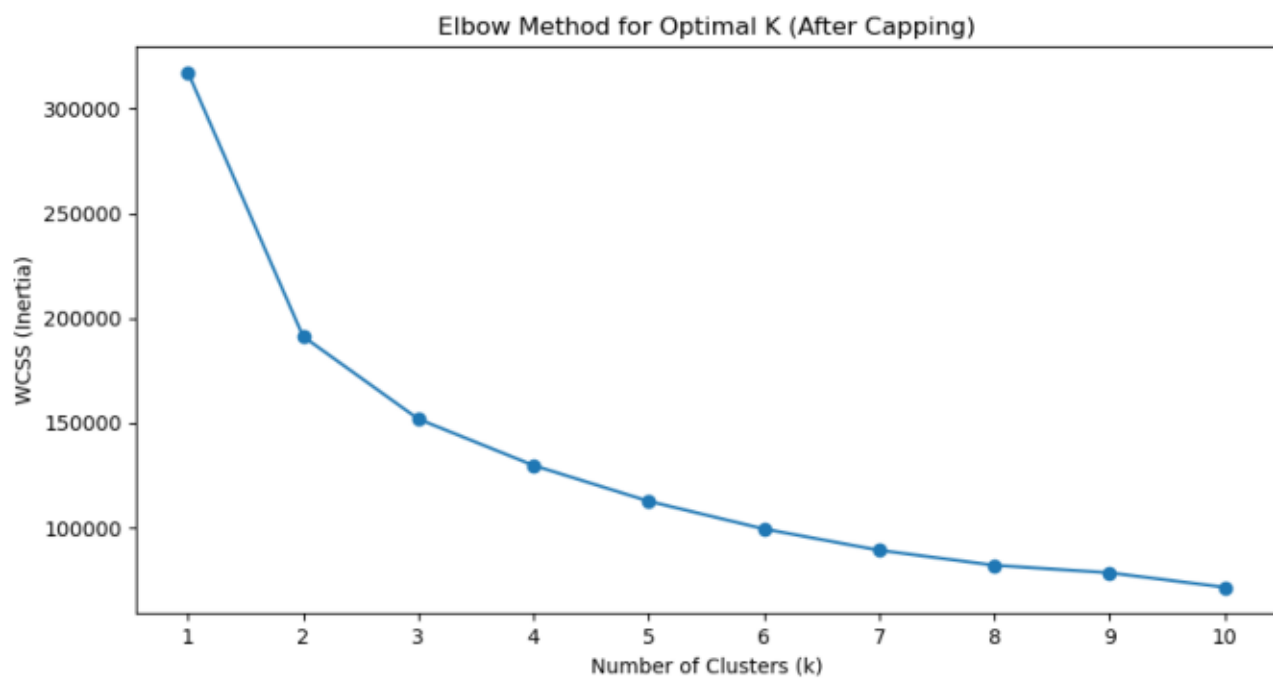
Payment Type Distribution — credit card dominates



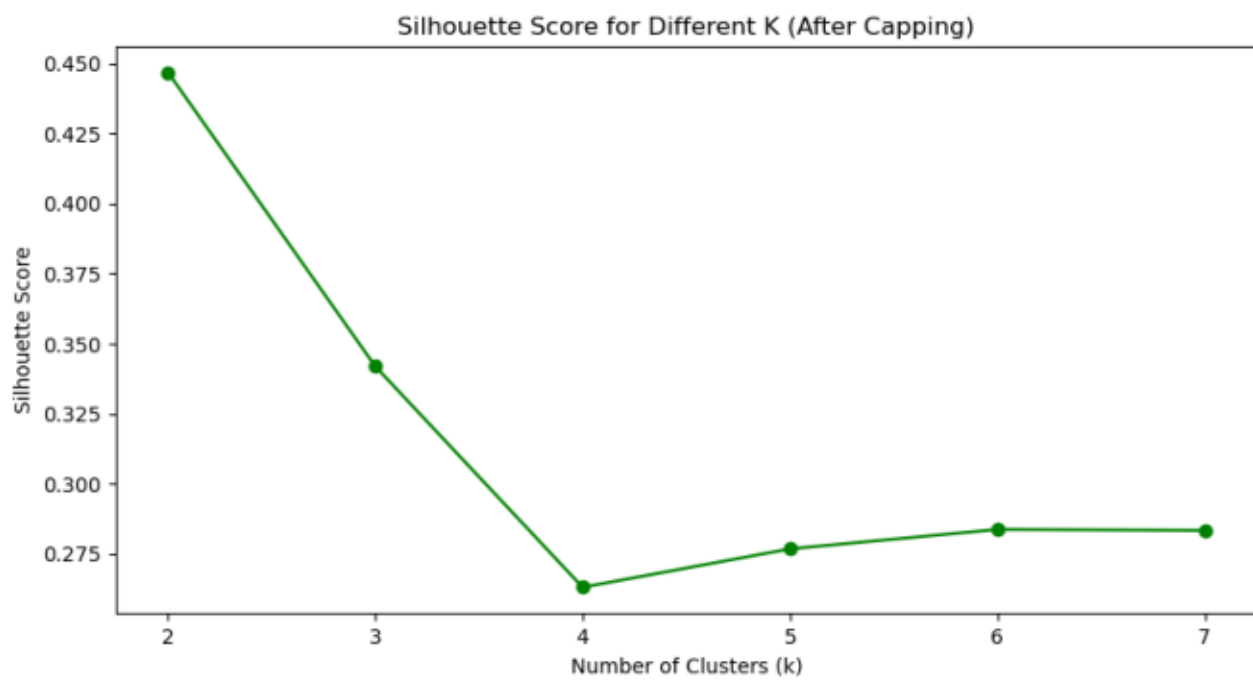
Orders by Customer State — São Paulo leads by a wide margin



RFM Boxplots — Monetary and Frequency show heavy outliers



Elbow Method (after outlier capping) — supports k=4



Silhouette Scores (after capping) — balanced clusters at k=4